

ARYAN BABURAJAN

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EXPERIENCE

Frappe.io Intern

Jan – Apr 2026

- Wrote a proof-of-concept Gunicorn worker using Python 3.14 Subinterpreters, enabling true concurrent parallel execution beyond the limitations imposed by the GIL.
- Integrated Bloomberg's Memray profiler into Frappe's web dashboard to support memory and performance analysis.
- Extended Redis Queue's WorkerPool to allow workers to process multiple queues concurrently, improving memory usage by 32%.

PROJECTS

Ducktape Creator

2021 – 2025

- Built an open-source C++ 3D game engine with an Entity Component System architecture using EnTT, attaining 141+ GitHub stars and 28 forks.
- Implemented real-time rendering in OpenGL, with prior renderer implementations in Vulkan.
- Supports native C++ DLL-based scripting and an ImGui-based editor for in-engine debugging and content creation workflows.

Attention Is All You Need Research

2026 - Present

- Implemented the original Transformer architecture from scratch in Python/PyTorch, including multi-head attention, sinusoidal positional embeddings, encoder-decoder layers, and feed-forward networks.
- Built a training and text-generation pipeline and experimented with training the model on a personal Discord message dataset.
- Implemented both encoder-decoder and decoder-only Transformer configurations with autoregressive text generation.
- SerenityOS - Improved Unicode support across applications and migrated them to SerenityOS's GML markup language.
- Mirage - Product Hunt-style platform built with Next.js, React, MongoDB, Kinde Auth, and Resend for email newsletters.
- X-Embed - Built a Twitter embedding service that generates perfect screenshots from URLs.
- Quaviz - Built an interactive 3D visualizer for comparing the scale of objects.

ACHIEVEMENTS

- 1st Place (₹15K) - Google Developer Groups Physical AI Hackathon
- Game Development Workshops - Held three workshops for 20+, 50+, and 100+ students on game development using p5.js, Unity, and Three.js.

TECHNICAL SKILLS

- Languages: C++, C, Rust, Python, TypeScript, JavaScript, C#
- Web: React, Next.js, Nest.js, Express.js, Node.js, WebSockets
- Graphics: OpenGL, Vulkan, EnTT, Dear ImGui, Unity, Blender
- Tooling: Git, CMake, Linux, MySQL, MongoDB, Netlify, Vercel, Heroku

EDUCATION

Christ College of Engineering

Bachelor of Technology in Computer Science and Engineering

2025 – Present
Kerala, India